

TO: Members of Congress

FROM: Wes Krikorian

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SUBJECT: Mandatory Disclosure of Validation and Demographic Data for FDA-Cleared AI/ML Medical Devices

1. Executive Summary

Over 1,451 AI/ML-enabled medical devices have been cleared by the FDA. A patient in a hospital today may be diagnosed, triaged, or routed to care by one of these devices without any reliable way to know whether it was ever tested on someone who looks like them.

This is not a hypothetical concern. The research literature documents, repeatedly, that AI medical tools trained on narrow populations perform worse on everyone else, sometimes with life-altering consequences. A widely used commercial care management algorithm was found to systematically underestimate the health needs of Black patients across hundreds of US health systems. Dermatology AI tools trained almost exclusively on lighter skin tones may fail to detect melanoma in patients with darker skin, with no requirement to say so. Pulse oximeters, the devices clipped to the finger of every hospitalized patient, were found during COVID-19 to produce readings so inaccurate in Black and Hispanic patients that care was delayed and medications were withheld. In each case, the problem was knowable. What was missing was any requirement to disclose it.

And yet the voluntary disclosure framework has failed: despite FDA guidance in place since 2021, fewer than one in four cleared AI devices reports any demographic information about its validation data.

The ask: Amend 21 CFR 807.92 to make the FDA's "Model Card" mandatory for any AI/ML-enabled device cleared via 510(k). If FDA does not act, amend FD&C Act § 510 by statute. This is a targeted, low-burden fix requiring structured public reporting of data manufacturers already collect internally.

Key data points:

- Over 1,451 AI/ML medical devices cleared by FDA, approximately 95% via the 510(k) pathway
- Only 23.7% of cleared AI devices report any dataset demographic data (Mehta et al., *npj Digital Medicine*, Nov. 2025)
- Only 15.5% of devices cleared in 2024 reported race or ethnicity of validation cohorts
- 30% of cleared AI devices report essentially nothing about their development or validation

2. Background & Problem Statement

The disclosure gap. 21 CFR 807.92 governs 510(k) summary content. It requires submitter information, device name, predicate ID, intended use, and clinical test subject descriptions, but does not require training-data demographics for AI/ML models. Because approximately 95% of AI/ML devices are cleared via 510(k), the public 510(k) summary is the primary, and often only, document patients and clinicians can access. Clearance turns on "substantial equivalence" to a predicate device, not demonstrated performance in a defined patient population. The disclosure gap is therefore structural, not incidental. Predicate chains compound the problem: one AI device cleared based on another, based on another, with opacity at every link.

The clinical stakes. The three examples below represent different clinical settings, different device types, and different mechanisms of harm. What they share is a common root: the population used to train or calibrate the tool did not reflect the population on whom it was deployed, and no one was required to say so.

Across hundreds of US health systems, a widely used commercial algorithm was quietly determining which patients were sick enough to be enrolled in care management programs (Obermeyer et al., *Science*, 2019). The algorithm had a flaw embedded in its design: it used health care costs as a proxy for health needs. Because structural inequities in access to care mean that less money is spent on Black patients with the same medical needs as White patients, the model read lower spending as lower need. At any given risk score, Black patients flagged by the algorithm were considerably sicker than their White counterparts, yet they were ranked lower priority. Correcting the disparity would have increased the share of Black patients receiving additional care from 17.7% to 46.5%, meaning the algorithm was effectively cutting access to care management for Black patients by more than half. When researchers and the manufacturer worked together to fix the underlying proxy, they reduced racial bias in outcomes by 84%. The solution was available. What was missing was the knowledge that the problem existed.

The same pattern appears in dermatology, one of the most active areas for AI development in medicine (Daneshjou et al., *JAMA Dermatology*, 2021). A scoping review of 70 studies examining AI algorithms for skin disease found that the datasets used to build these tools almost never described who the patients were. Only 14 of 70 studies reported any race or ethnicity information. Only 7 reported any information on patient skin tone. Of those 7, four reported having no images of the darkest skin types — Fitzpatrick V and VI — whatsoever. The three studies that did include darker skin tones had representation so low as to be clinically meaningless: one had 3.2% type V and 0.3% type VI images in its training data. An AI tool trained almost entirely on lighter skin tones may simply not recognize melanoma reliably in patients with darker skin, and the clinician using it has no way of knowing. The authors found no reporting guidelines specific to dermatology AI existed and called for their development.

Of all the medical devices shaped by this problem, the pulse oximeter may be the most consequential simply because of how ubiquitous it is (Fawzy et al., *JAMA Internal Medicine*, 2022; Sjoding et al., *NEJM*, 2020). Clipped to the finger of virtually every hospitalized patient, it measures blood oxygen saturation and during COVID-19 became a gatekeeper: federal guidelines directed patients to seek emergency care only if their reading fell below 90%, and Medicare reimbursement rules tied to that same threshold. Pulse oximeters work by passing light through the skin, and their accuracy is affected by pigmentation. When the devices were originally developed, the validation populations were not diverse. The FDA's premarket testing standard, unchanged from 2013, required manufacturers to include just two darkly pigmented subjects, the same inadequate ratio researchers had flagged as insufficient for detecting bias as far back as 1991. During the pandemic, Black patients were found to be approximately three times as likely as White patients to have a significant discrepancy between their pulse oximeter reading and their actual blood oxygen level, resulting in a documented systemic failure: delayed care and in some cases denial of medications for Black and Hispanic patients. The clinicians treating those patients had no reason to distrust the device. It was FDA-cleared. There was nothing in the public record telling them it had not been adequately tested across skin tones.

Why voluntary disclosure has failed. The empirical record is unambiguous. Guidance has been in place since 2021, yet the majority of cleared AI devices remain opaque about their training populations. Average transparency scores sit at 3.3 out of 17 possible points; 30% of cleared devices score zero. Manufacturers face a structural competitive disincentive to disclose unilaterally, and a mandate is the only mechanism that solves that collective-action problem.

Precedents supporting this ask. The ONC HTI-1 Final Rule (2024) requires "Decision Support Intervention" source attributes including underlying data information, the strongest existing HHS precedent for mandatory algorithmic transparency. FDA already requires clinical trial demographic reporting for drugs under FDORA 2022; there is no principled basis for treating AI devices differently. The EU AI Act imposes hard documentation obligations for medical device AI beginning August 2026 to 2027; U.S. manufacturers will produce this documentation for European regulators regardless. The only question is whether U.S. patients can see it.

3. Policy Options

Option A — FDA rulemaking (amend 21 CFR 807.92): Require a standardized Model Card in the public 510(k) summary for all AI/ML-enabled devices. This uses existing FDA authority with no new statute needed, and FDA already signaled intent via its January 2025 draft guidance. Notice-and-comment rulemaking takes 2 to 4 years and is vulnerable to administration change if not completed.

Option B — Statutory amendment (FD&C Act § 510): Congress directly requires Model Card disclosure in law. This is the strongest and most durable path, not dependent on FDA rulemaking

pace or priorities. It requires Congressional action and is best pursued through MDUFA reauthorization or an AI omnibus bill.

Option C — Finalize January 2025 draft guidance with mandatory language: Direct FDA to finalize its existing draft with binding rather than recommendatory language. This is the fastest path and anchors the ask to FDA's own prior work, but guidance remains technically non-binding even when finalized.

Option D — Special controls: FDA imposes Model Card disclosure as a special control for AI/ML device categories. This can be deployed class-by-class without waiting for broad rulemaking but is piecemeal and may not achieve universal coverage.

4. Recommendation

Primary ask: Option A with Option B as explicit fallback.

Congress should direct FDA to issue a final rule amending 21 CFR 807.92 to require, for any device incorporating an AI/ML-enabled software function, a standardized Model Card in the public 510(k) summary disclosing:

- Training and validation dataset sources, sizes, and time periods
- Demographic composition of training and validation populations (age, sex, race/ethnicity using the current OMB 2024 six-category standard, and geographic origin)
- Intended-use population
- Known performance differences across subpopulations
- Limitations on generalizability

If FDA does not initiate rulemaking within 12 months of Congressional direction, Congress should amend FD&C Act § 510 to require it directly. An 18-month implementation deadline from enactment gives manufacturers adequate time to comply.

Addressing counterarguments:

"Trade secrets" — The Model Card discloses aggregate training-data characteristics, not the data itself or model weights. This is a nutrition label, not a recipe.

"Burden on small developers" — Manufacturers already produce this data internally for validation purposes. The mandate requires structured public reporting of existing information, not new data collection.

"Redundant with existing guidance" — Guidance has been in place since 2021 and voluntary uptake remains under 25%. The empirical record refutes redundancy.

"EU compliance is enough" — The EU AI Act covers EU-market devices only. U.S. patients are not protected by it.

5. Implementation Considerations

Timeline. Congressional direction to FDA should ideally be tied to the MDUFA reauthorization cycle. FDA notice-and-comment rulemaking will take 18 to 24 months, followed by an 18-month manufacturer compliance window after the final rule. Fallback statutory language can be drafted as short, single-purpose legislative text attachable to a larger vehicle.

Stakeholders. FDA's CDRH Digital Health Center of Excellence is already moving in this direction and is receptive to a Congressional nudge. AdvaMed and device manufacturers will likely oppose; trade-secret and burden concerns should be addressed directly in outreach. The American College of Radiology and other clinician societies are key validators, given that approximately 76% of cleared AI devices are in radiology. Patient advocacy organizations including Light Collective and Patient Privacy Rights are likely allies.

Risks. Administration change could stall FDA rulemaking, which is why the statutory fallback is essential. Industry litigation challenging FDA's authority to require a Model Card via guidance rather than rulemaking is plausible; the rulemaking path is more legally defensible. Scope should be kept narrow: pre-market disclosure only. Post-market demographic performance reporting is a logical next step but a separate fight.

Parallel actions. File a Citizen Petition (21 CFR 10.30) requesting FDA initiate rulemaking to amend 807.92, which forces a formal written FDA response and creates a public record. Submit comments if the January 2025 draft guidance is still open or reopened. A 700-word op-ed in STAT News or Health Affairs Forefront with a clinician co-author, using a patient-facing example as the lede, can drive staff attention and broaden the coalition.

6. Conclusion

FDA has already identified the problem and recommended the solution. Mandatory disclosure is the logical next step, not a new regulatory burden. Three examples spanning a commercial care management algorithm, a dermatology diagnostic tool, and a device found in every hospital room tell the same story: when the demographics of a validation population go undisclosed, clinicians cannot know the limits of the tool they are using, and patients bear the consequences.

The empirical record is clear: voluntary compliance has plateaued at under 25%, and a mandate is the only mechanism that closes the gap. This is a narrow, technically straightforward fix requiring structured public reporting of data manufacturers already hold, with significant patient safety and health equity upside.

The U.S. risks falling behind. EU AI Act documentation requirements take effect in 2026 to 2027, and U.S. manufacturers will produce this data for European regulators regardless. The question is only whether American patients receive the same transparency protections their European counterparts will.

Call to action: Direct FDA to initiate rulemaking to amend 21 CFR 807.92 within 12 months, with statutory fallback language under FD&C Act § 510 ready if FDA does not act.